python-ML-Aishwarya / _Supervised_learning.ipynb

Aishwarya-eee Add files via upload 6b9c00a · 2 weeks ago

92 lines (92 loc) · 50.5 KB

Preview Code Blame Raw

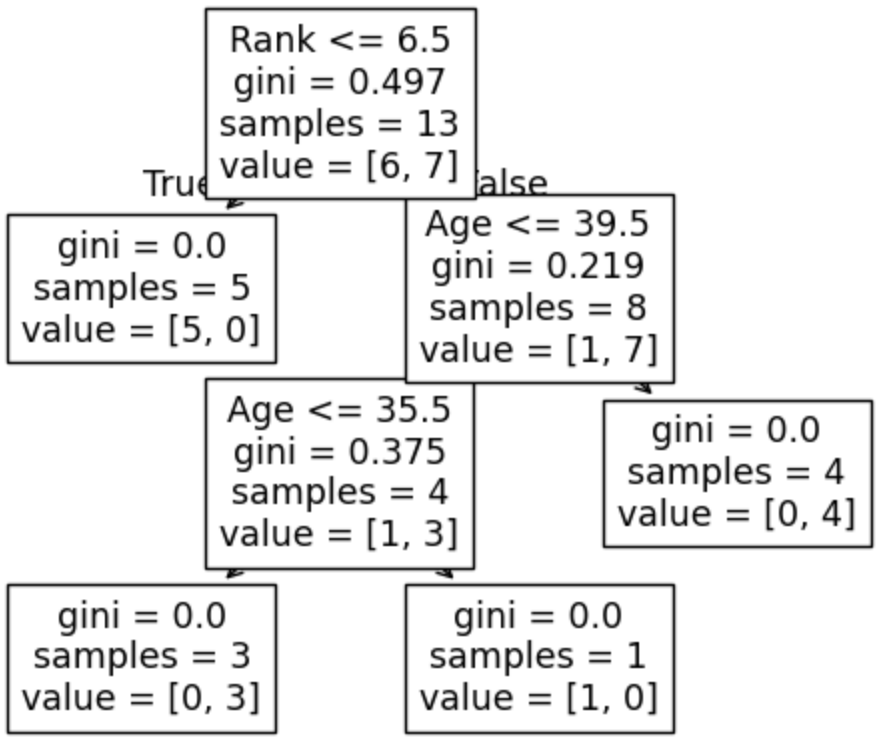
In [1]:

```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn import tree
from sklearn.tree import DecisionTreeClassifier
import matplotlib.pyplot as plt

df = pd.DataFrame({'Age': [36,42,23,52,43,44,66,35,52,35,24,18,45],
'Experience': [10,12,4,4,21,14,3,14,13,5,3,3,9],
'Rank': [9,4,6,4,8,5,7,9,7,9,5,7,9],
'Nationality': ['UK', 'USA','N','USA','USA', 'UK','N','UK','N','N',
                'USA', 'UK', 'UK'],
'Go': ['NO', 'NO', 'NO', 'NO', 'YES','NO', 'YES','YES',
       'YES', 'YES', 'NO', 'YES','YES'] })
print(df)

d = {'UK': 0, 'USA': 1, 'N': 2}
df['Nationality'] = df['Nationality'].map(d)
d = {'YES': 1, 'NO': 0}
df['Go'] = df['Go'].map(d)

features = ['Age', 'Experience', 'Rank', 'Nationality']
X = df[features]
y = df['Go']

dtree = DecisionTreeClassifier()
dtree = dtree.fit(X,y)
tree.plot_tree(dtree, feature_names=features)
plt.show()
```

```
    Age  Experience  Rank Nationality   Go
0    36          10     9          UK   NO
1    42          12     4         USA   NO
2    23           4     6           N   NO
3    52           4     4         USA   NO
4    43          21     8         USA  YES
5    44          14     5          UK   NO
6    66           3     7           N  YES
7    35          14     9          UK  YES
8    52          13     7           N  YES
9    35           5     9           N  YES
10   24           3     5         USA   NO
11   18           3     7          UK  YES
12   45           9     9          UK  YES
```